

MegaMart System Specification

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System Overview

MegaMart Online Marketplace is a comprehensive e-commerce platform designed on both C2C and B2C models, acting as a direct bridge between **Buyers** and **Sellers**. The system aims to provide a transparent, secure, and intelligent online trading environment.

The key differences between MegaMart and traditional e-commerce platforms lie in the application of advanced technologies:

1. **Intelligent Shopping Experience:** Integrates an AI Virtual Assistant (RAG Chatbot) utilizing OpenAI and a Vector Database (pg vector) to provide accurate product recommendations based on real-world context and customer needs.
2. **Real-time Interaction:** Supports direct real-time chat between Buyers and Sellers, along with an instant notification system triggered by order status changes, powered by WebSocket technology.
3. **Comprehensive Management Ecosystem:** Provides powerful tools ranging from store management (Seller Dashboard) with service packages, flash sales, and vouchers, to deep moderation, authorization, and revenue reporting systems for the Administration Board (Admin Dashboard).
4. **Sustainable Architecture:** The system is built on a Modular Monolith foundation combined with Domain-Driven Design (DDD), ensuring the codebase is easy to maintain, feature-extensible, and ready to transition to Microservices in the future when traffic scales.

Below is the detailed specification document covering the Actors, Core Features, and Workflows of the MegaMart system.

1. Actor

1.1 Unauthenticated User (Guest)

- Access the homepage.
 - View the product catalog.
 - Search and filter products.
 - View product details.
 - Register an account (Registration/login is required to buy or sell).
-

1.2 Authenticated Customer (Buyer)

- All privileges of a Guest.
 - Register / Login / Logout.
 - Add products to the shopping cart.
 - Checkout and place orders.
 - View order history.
 - Update personal information.
 - Review/Rate products and sellers (optional).
 - Send contact requests/messages to sellers.
 - Report inappropriate or violating products.
-

1.3 Seller

- Post products (CRUD operations).
- Manage the product inventory.
- Manage orders (view customer details, update order status).
- Manage messages/inquiries from Buyers.

- View reviews and revenue statistics.
-

1.4 System Admin

- Manage user accounts (approve, ban, delete).
 - Manage authorization/roles (Buyer / Seller / Admin).
 - View system-wide statistics (users, products, orders, revenue).
 - Manage system configurations (Payment, Storage, etc.) (optional).
 - Set and adjust the commission percentage for each order.
 - Create / Update / Delete service packages for Sellers.
 - Create system-wide vouchers.
-

1.5 Content Admin

- Approve products before they become visible on the marketplace.
 - Manage product reports (spam, violations).
 - Manage reviews and comments (if any).
 - Manage and control vouchers (approve, toggle on/off, track usage).
 - View simple content statistics (Number of pending products, reported products, violation count per month, active vouchers, hidden/removed reviews, product approval rate).
-

1.6 External Actors (Optional)

- **Payment Gateway (mock):** Processes online order payments.
- **Delivery Service (mock):** Manages shipping statuses (simulates shipping and updates order status).
- **Email/SMS Service (Mail Gateway - SendGrid, Twilio):** Sends system notifications.
- **Chatbot AI Service (OpenAI API, Rasa, Dialogflow):** Provides automated user support.

- **Cloud Storage (Firebase, AWS S3):** Stores images and documents.
-

2. Core Features

1. Direct Chat (Buyer ↔ Seller)

- A built-in mini-messenger or a "Contact Seller" form.
- Be implemented via WebSockets (real-time) or by simply storing messages in the database.

2. Wishlist

- Users can "like" or "save" products to purchase later.
- Implemented via a wishlist table (user_id, product_id).

3. Voucher / Promotions

- Sellers can create discount codes for their own products.
- Buyers enter the code during checkout to receive discounts.

4. Notifications

- Triggered when an order is placed, a product is approved, or a new promotion is available.
- Real-time (WebSocket) or displayed statically in the User Profile.

5. Review & Comment

- Buyers can rate and review a product after receiving their order.
- Enhances the "social" aspect of the platform.

6. Advanced Admin Role Management

- Separation of duties between System Admin and Content Admin.
- Can potentially include a Finance Admin (specifically for tracking transactions).

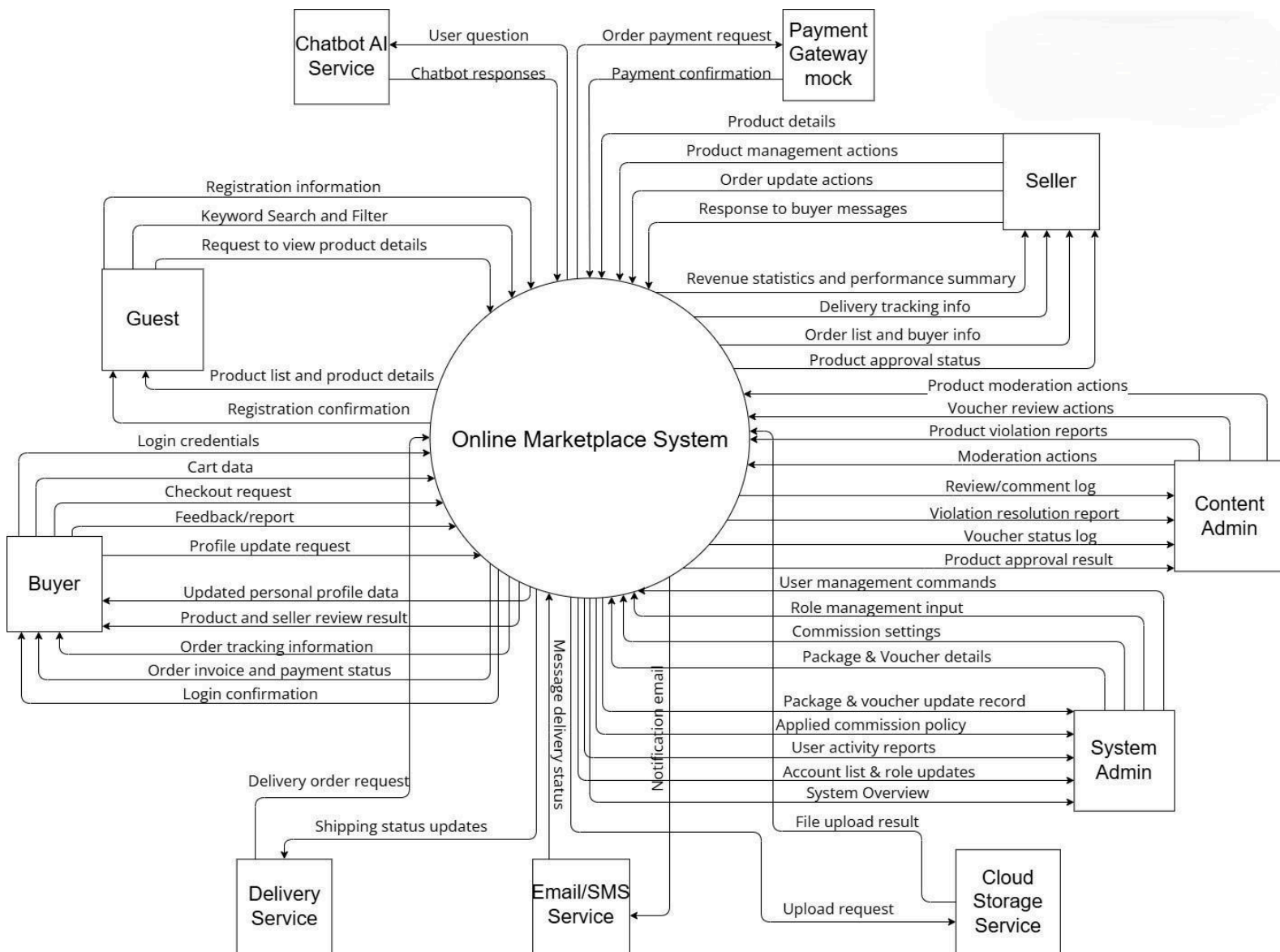
7. Product Reporting

- Buyers can report violating products (counterfeits, spam).
- Content Admins will handle these reports.

8. Core Trading (Buying & Selling)

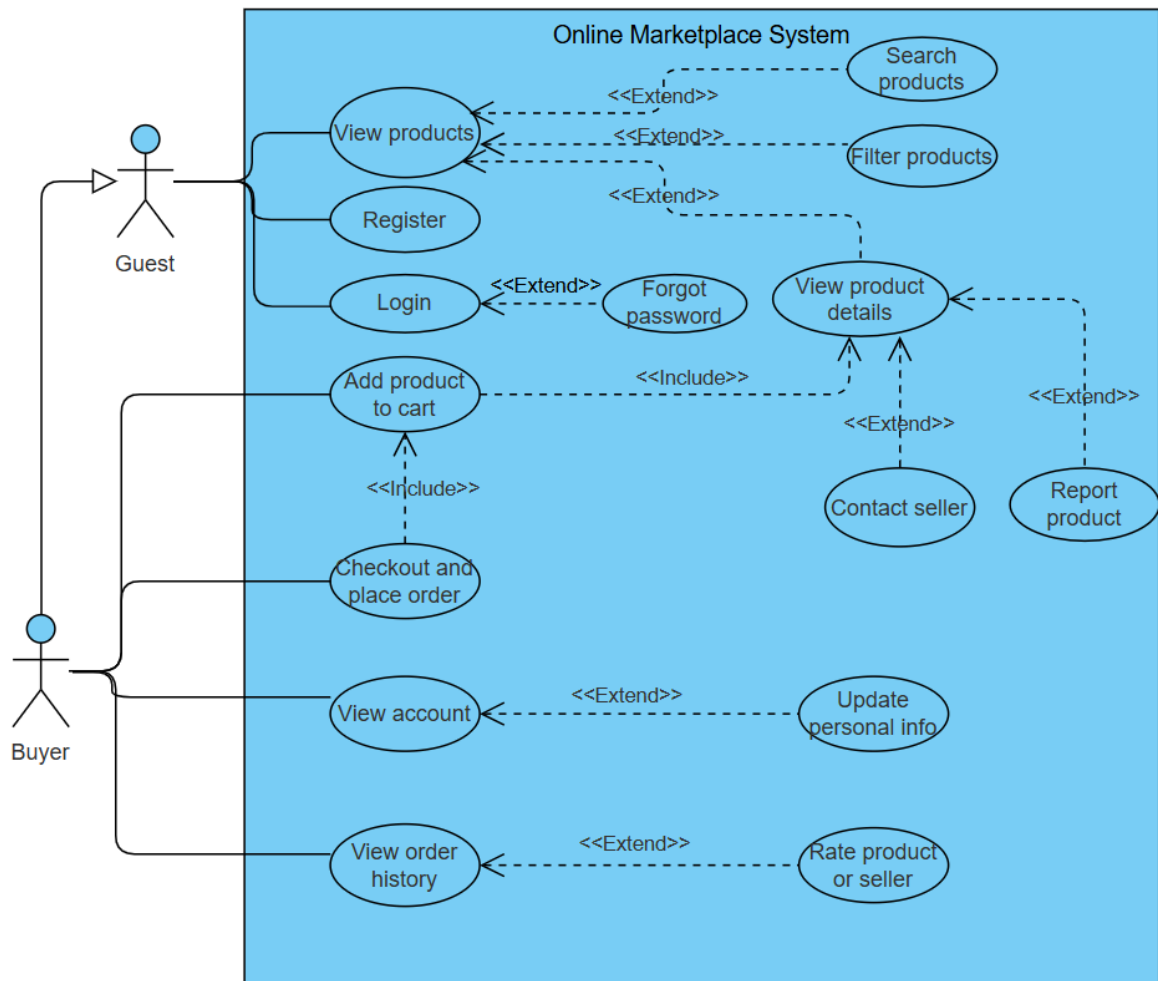
- Buyers can place orders based on their needs.
 - Sellers can perform CRUD operations on the items they sell.
-

3. Context Diagram

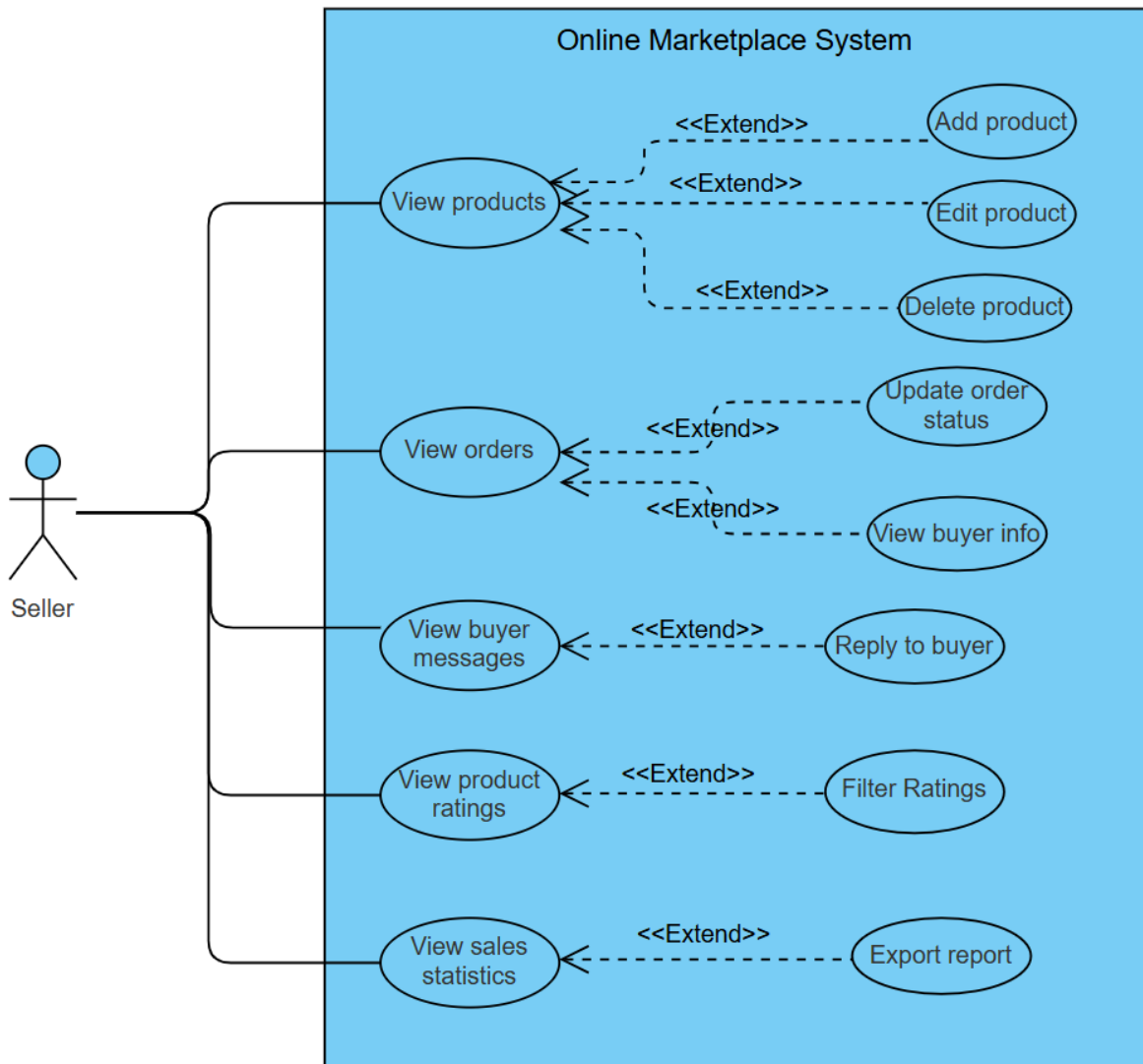


4. UseCase

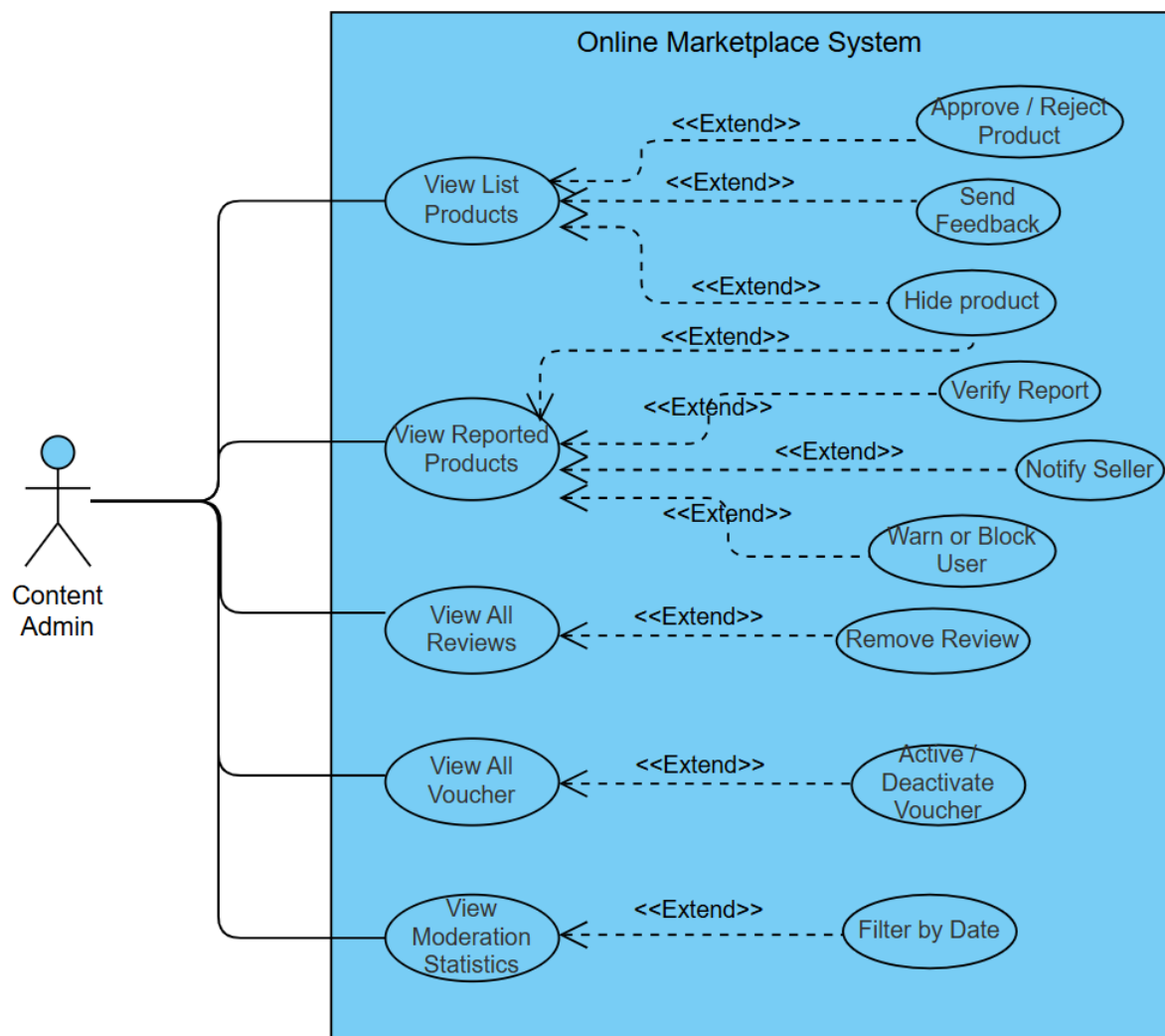
1. Guest và Buyer



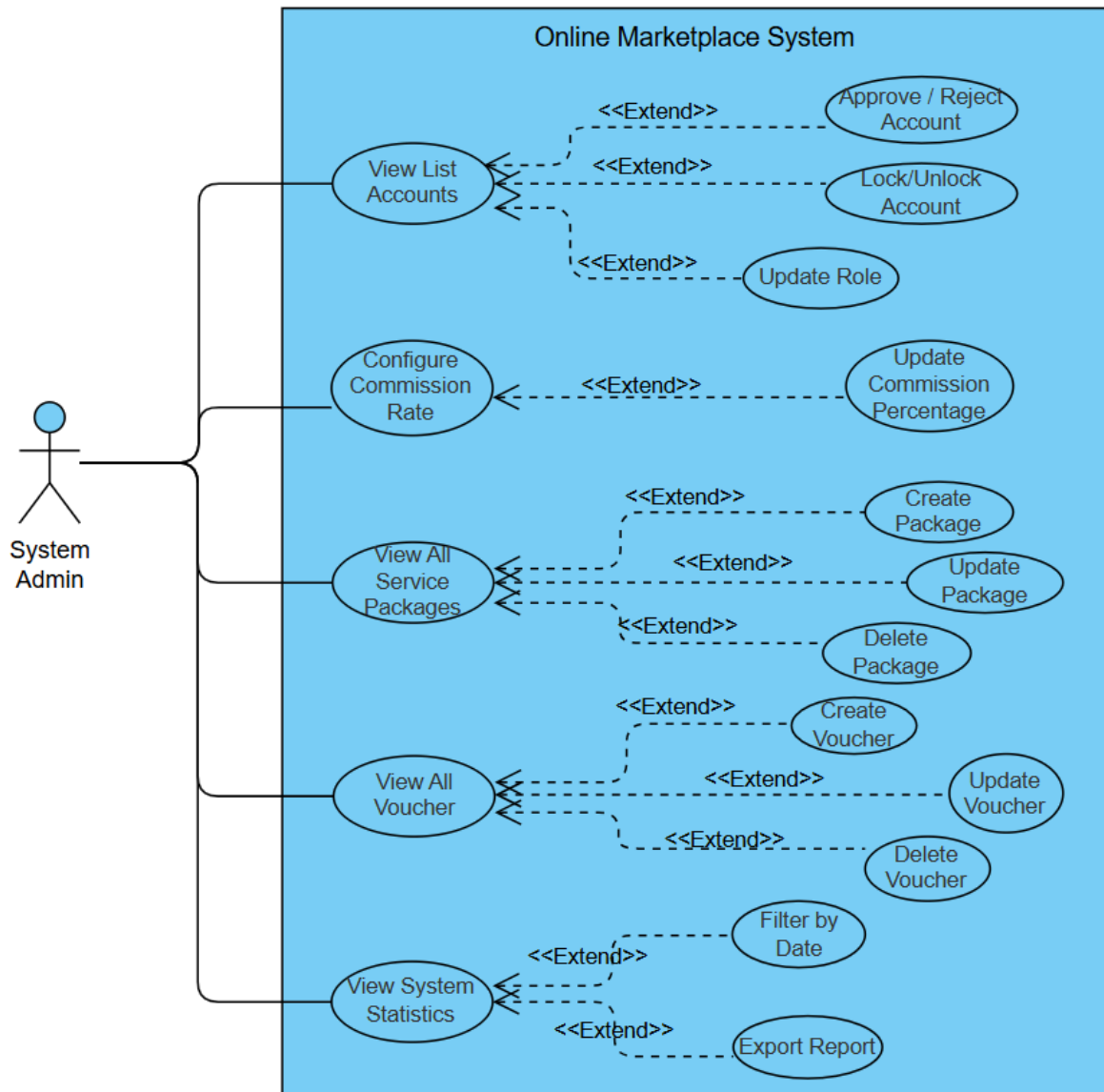
2. Seller



3. Content Admin



4. System Admin



5. Database



6. Workflow

6.1. Buyer Workflow

WF 1: User Registration & Login (Email / Google)

Actors: Buyer, System, Google OAuth

Flow:

1. Buyer chooses to register or login.
2. If using Google Login → redirect to Google OAuth → verify → return user info
3. If using Email → system validates email/password and OTP if required.
4. On success → user session created, redirect to homepage.

WF 2: Browse & Search Products

Actors: Buyer, System

Flow:

1. Buyer enters search keywords or browses categories.
 2. System filters and displays products with pagination.
 3. Buyer can sort or apply filters (price, rating, brand, etc.).
 4. Buyer selects a product to view details.
-

WF 3: Shopping Cart Management

Actors: Buyer, System

Flow:

1. Buyer adds/removes items to/from the cart.
2. Buyer updates quantity or views total price.
3. System validates stock and price changes in real time.
4. Buyer proceeds to checkout.

WF 4: Checkout & Place Order (with Promo Code)

Actors: Buyer, System, Payment Gateway, Seller

Flow:

1. Buyer reviews cart and applies promo code (if any).
 2. System calculates final total after discount.
 3. Buyer selects payment method (Online / COD).
 4. Payment Gateway processes online transactions.
 5. System creates order → notifies Seller.
 6. Admin receives commission (if applicable).
-

WF 5: Order Tracking

Actors: Buyer, Seller, System

Flow:

1. Buyer opens “My Orders” page.
 2. System displays status (Pending → Processing → Shipped → Delivered).
 3. Buyer can confirm delivery or request return/refund.
-

WF 6: Review & Report Products

Actors: Buyer, System, Admin, Seller

Flow:

1. Buyer writes a product review after order completion.
 2. System verifies the order before posting.
 3. Buyer can also report a product (fake, inappropriate, etc.).
 4. Admin receives report → review and take action.
-

WF 7: Chat with Seller

Actors: Buyer, Seller, System

Flow:

1. Buyer opens chat window on product page or order detail.
 2. Message is stored and synced in real time.
 3. Both sides can communicate directly via in-app chat.
-

WF 8: Wishlist Management

Actors: Buyer, System

Flow:

1. Buyer adds/removes products to/from Wishlist.
 2. System stores wishlist in user account.
 3. Buyer can access or move items to cart anytime.
-

WF 9: Notification Flow

Actors: System, Buyer

Flow:

1. System triggers notifications (order updates, promo, chat, etc.).
 2. Notifications appear in-app and optionally via email.
 3. Buyer can mark as read or clear all.
-

6.2. Seller Workflow

WF 10: Login & Store Management

Actors: Seller, System

Flow:

1. Seller logs into Seller Dashboard.
 2. System authenticates account and loads store info.
 3. Seller manages profile, products, and orders.
-

WF 11: Product Management (Add / Edit / Delete)

Actors: Seller, System, Admin

Flow:

1. Seller adds/edits product → set price, stock, description, images.
 2. Product status = Pending Approval.
 3. Admin reviews → Approve / Reject.
 4. Approved products become visible to Buyers.
-

WF 12: Order Management & Shipping Updates

Actors: Seller, Buyer, System

Flow:

1. Seller views new orders.
 2. Updates order status: Processing → Shipped → Delivered.
 3. System notifies Buyer of each update.
 4. Commission calculated when order completes.
-

WF 13: Flash Sale Registration

Actors: Seller, System, Admin

Flow:

1. Seller requests to join Flash Sale campaign.
2. System validates product eligibility.

3. Admin reviews and approves participation.
 4. Product becomes visible in Flash Sale list.
-

WF 14: Purchase & Use Seller Package

Actors: Seller, System, Payment Gateway, Admin

Flow:

1. Seller views available service packages.
 2. Chooses package → makes payment (Online / COD).
 3. System activates package and features (e.g., ad boost, analytics).
 4. Admin earns commission from package purchase.
-

WF 15: View & Reply to Reviews

Actors: Seller, System, Buyer

Flow:

1. Seller views product reviews from Buyers.
 2. Can reply publicly or report inappropriate content.
 3. System updates product rating and review history.
-

6.3. Admin Workflow

WF 16: Product Approval

Actors: Admin, System, Seller

Flow:

1. Admin views pending products.
 2. Reviews details → Approve or Reject.
 3. System updates status and notifies Seller.
-

WF 17: Promotion Management

Actors: Admin, System

Flow:

1. Admin creates and manages discount codes or global campaigns.
 2. System validates and applies promotions during checkout.
 3. Reports generated for promo performance.
-

WF 18: Flash Sale Management

Actors: Admin, System, Seller

Flow:

1. Admin creates Flash Sale events.
 2. Approves seller participation.
 3. Manages product lists and schedules.
 4. Ends or extends sale manually if needed.
-

WF 19: Service Package Management

Actors: Admin, System

Flow:

1. Admin defines available Seller Packages (price, duration, benefits).
 2. Monitors purchases and renewals.
 3. Updates or removes outdated packages.
-

WF 20: Complaint & Report Handling

Actors: Admin, Buyer, Seller, System

Flow:

1. Buyer submits complaint or report.
2. Admin reviews details and evidence.
3. Contacts involved Seller if necessary.

4. Takes action: Warning / Product removal / Account suspension.
5. System updates complaint status to "Resolved".

WF 21: User Management

Actors: Admin, System

Flow:

1. Admin views user list (Buyers, Sellers).
2. Changes status: Active / Banned / Suspended.
3. Assigns or removes roles and permissions.
4. Logs all actions in SystemLog.

WF 22: Send System Notifications

Actors: Admin, System

Flow:

1. Admin composes notification message.
2. Chooses target group (All, Sellers, Buyers, Specific users).
3. System broadcasts notifications.
4. Users receive in-app and email alerts.

WF 23: Commission Management (from Orders / Packages)

Actors: Admin, System, Seller

Flow:

1. Order Completed → System checks payment method.
 - Online: Admin keeps full payment → deducts 10% → transfers 90% to Seller.
 - COD: Seller collects cash → system records 10% debt → Seller pays commission later.
 2. Admin views total commission reports.
 3. System logs all commission events.
-

WF 24: System Log Monitoring

Actors: Admin, System

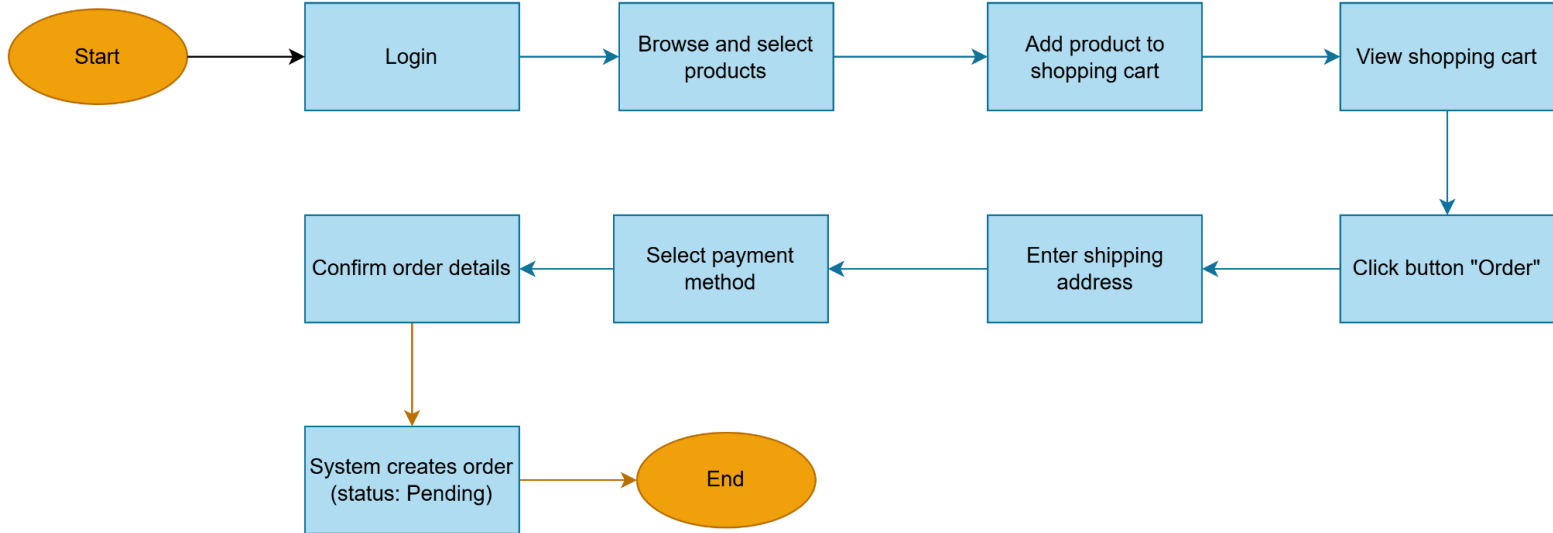
Flow:

1. Admin opens System Logs section.
2. Filters by user, action, module, or date.
3. Views detailed logs for each event.
4. Exports or clears logs based on retention policy.

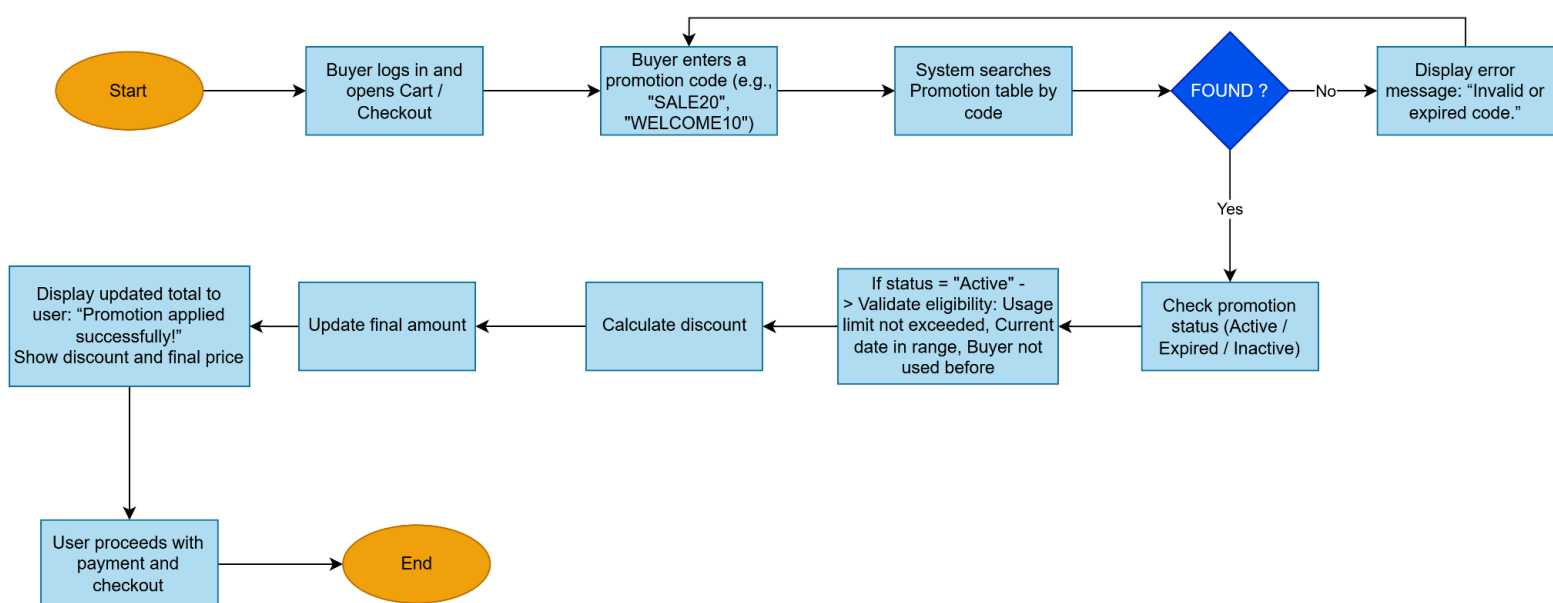
7. Demo

1. Checkout & Place Order (with Promo Code)

4.1. Place Order (Buyer)

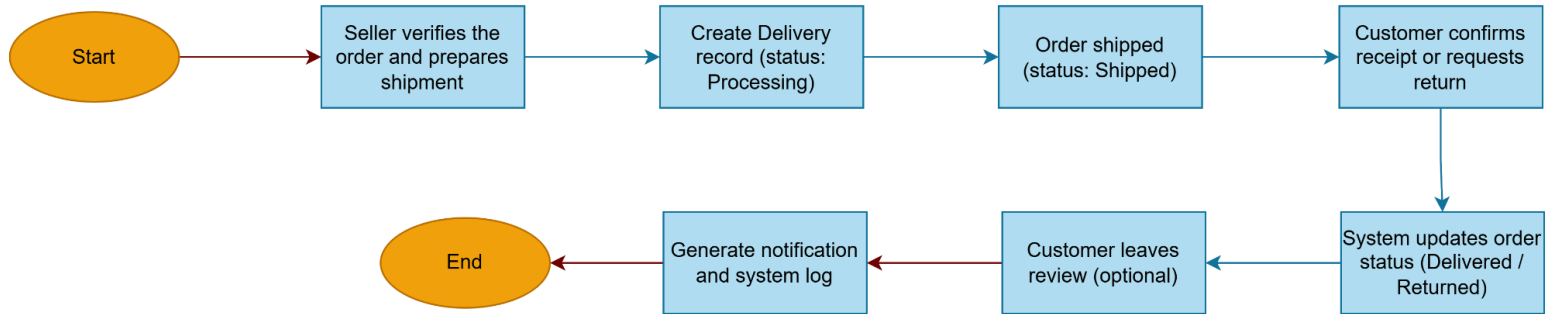


4.2. Apply Promotion Code (Buyer)



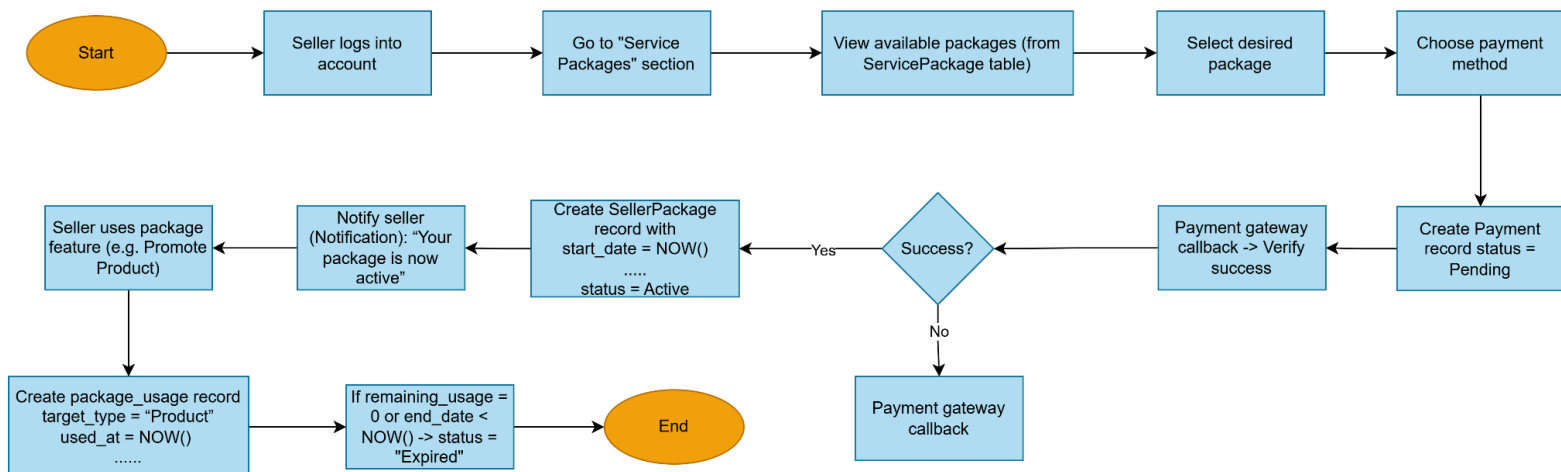
2. Order Management & Shipping Updates

12. Manage Order & Delivery Process (Seller)



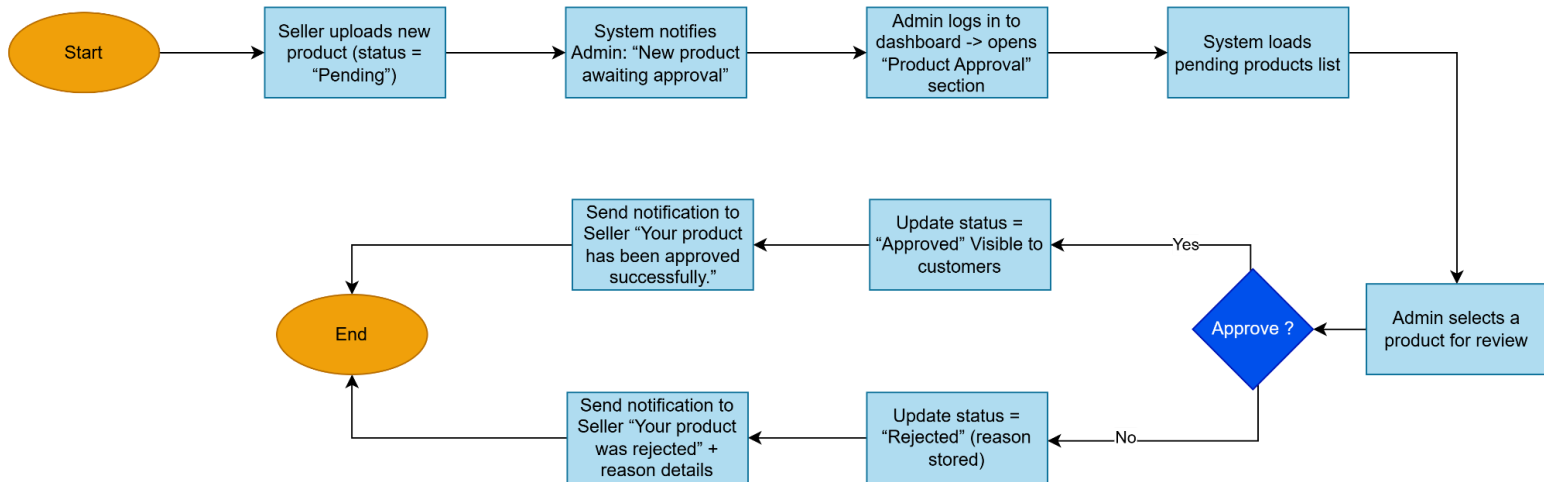
4. Purchase & Use Seller Package

14. Seller Service Package Flow (Seller)



3. Product Approval

16. Product Approval Flow (Admin)



5. Service Package Management

19. Service Package Management (Admin)

